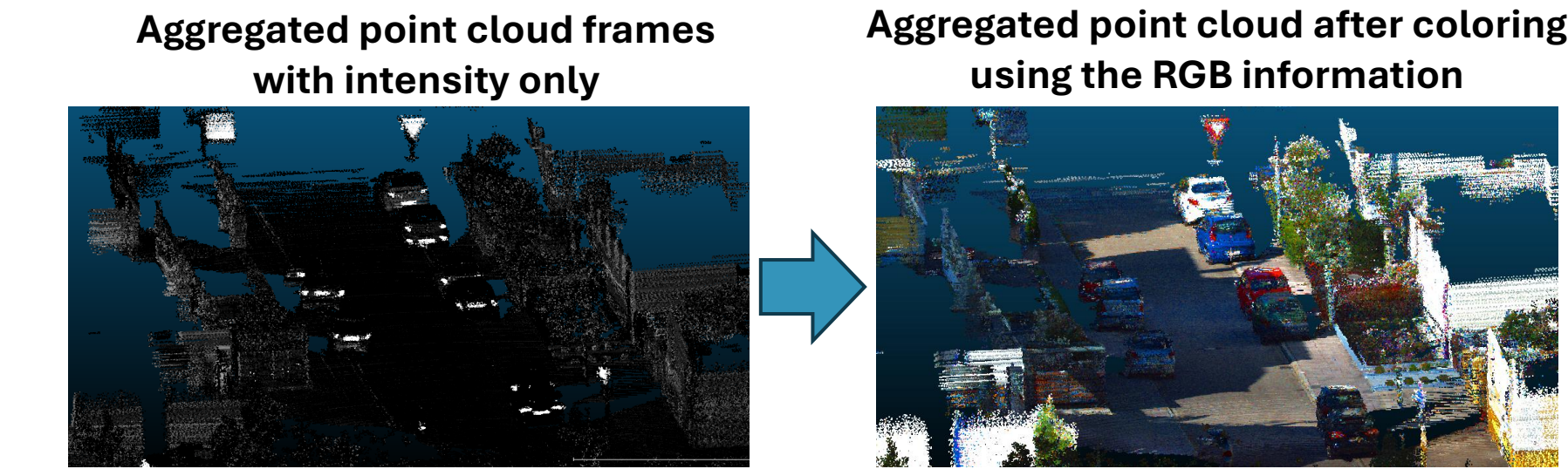


Motivation

- Camera-LiDAR fusion depends on **accurate extrinsic calibration**.
- Multi-camera platforms share one LiDAR, so camera extrinsics are **geometrically coupled**.
- Current learning-based methods calibrate each **camera-LiDAR pair independently**.
- This can produce individually plausible but **jointly inconsistent** calibrations.
- Goal: combine learned pairwise matching with rig-aware geometric refinement.**



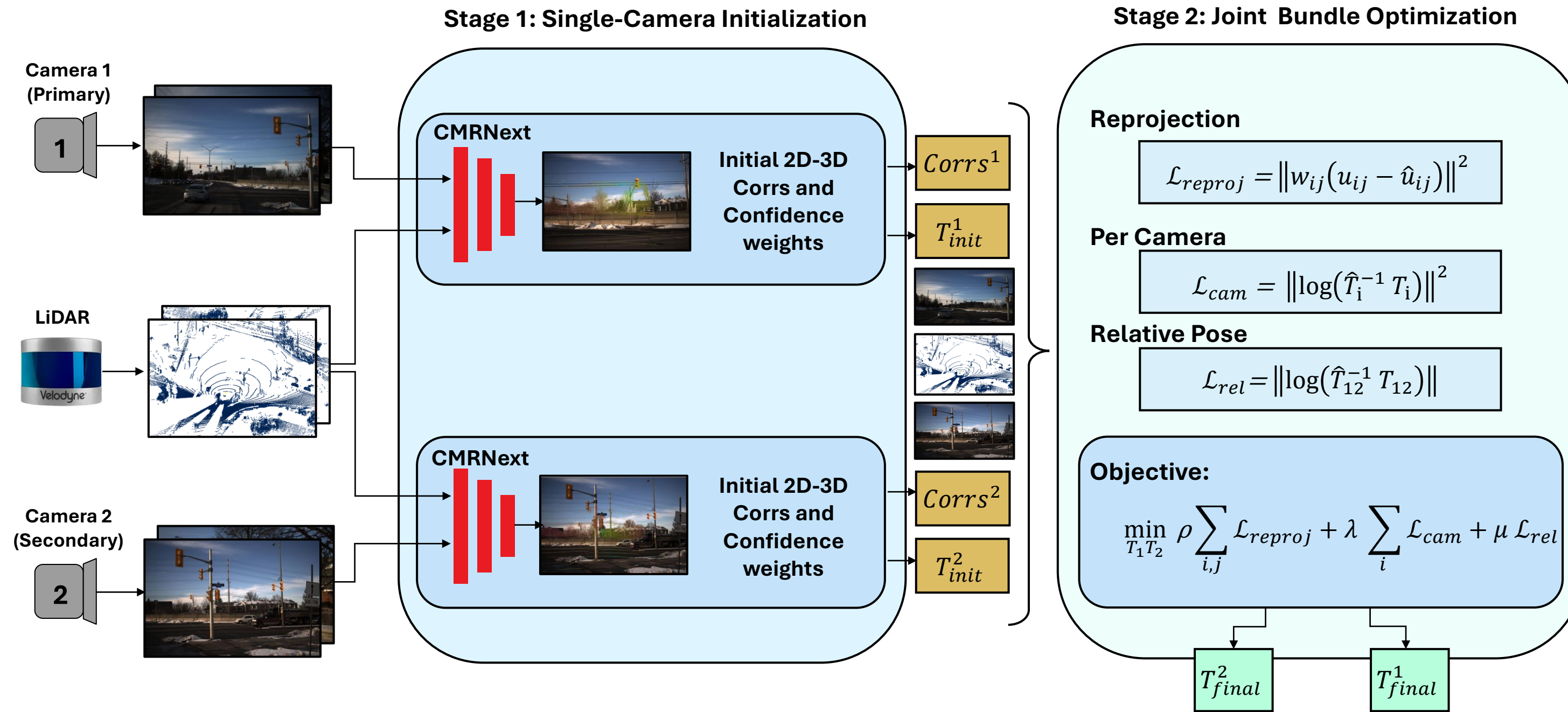
Key Contributions

- A two-stage joint multi-camera calibration framework.
- Learned-based single-camera initial calibration.
- Multi-frame bundle adjustment with reprojection, per-camera, and relative-pose priors.
- Improved per-camera accuracy and inter-camera consistency on both training dataset and zero-shot out-of-domain datasets.

Setup

Datasets	KITTI Sequence 00-01: stereo cameras + Velodyne HDL-64E, 4541 frames, in-domain of the CMRNext model. Walkley: 2 forward-facing cameras + Velodyne 128CH-Alpha Prime LiDAR, 226 frames, out-of-domain.
Settings	- Frame subsets N = 10 and 100. - Initial perturbation: +1.5 m/ +20°.
Metrics	- Rotation Error E_R (°). - Translation Error E_t (cm). - Inter-camera Consistency (IC-ΔR and IC-Δt).

Proposed Method



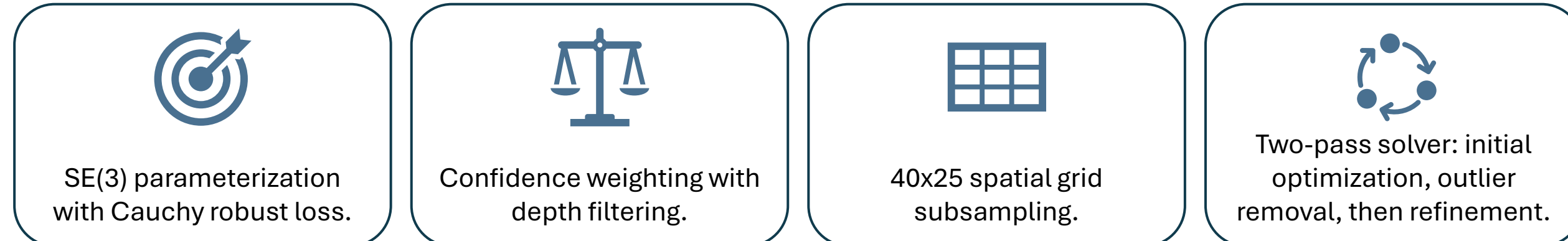
Stage 1: Learned Single-Camera Initialization

- For each synchronized image-point-cloud pair, CMRNext predicts dense optical flow between a rendered LiDAR depth map and the camera image.
- This yields confidence-weighted 2D-3D correspondences (Corrs) and a per-frame extrinsic estimate.
- Median aggregation across frames provides a robust initial extrinsic for each camera.

Stage 2: Joint Multi-Camera Bundle Adjustment

- Reprojection term aligns LiDAR points to observed image locations.
- Per-camera prior keeps each solution near its Stage-1 estimate.
- Relative-pose prior enforces consistency between cameras through the shared LiDAR frame.

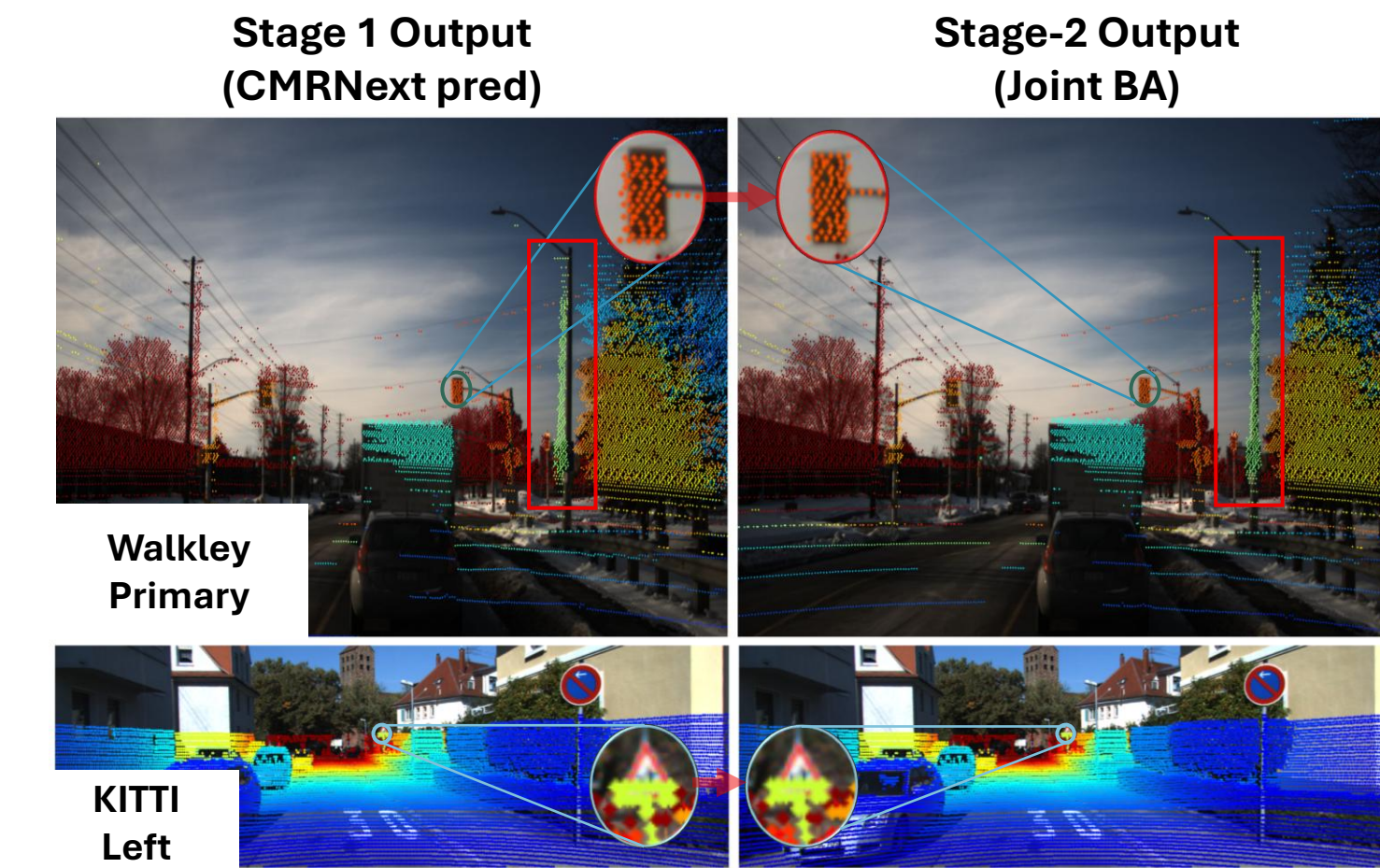
Optimization Details



Key Results

- KITTI Dataset:**
 - Achieved 0.89 cm translation error and 0.038° rotation error on the primary camera using 100 frames.
 - Improved Stage-1 CMRNext **from 2.63 cm / 0.187° to 0.89 cm / 0.038°**.
 - Reduced inter-camera rotational inconsistency from **0.206° to 0.037°**.
 - Shows that joint optimization improves both single-camera accuracy and multi-camera consistency.

KITTI 00-01	Left Camera (Primary)		Left Camera (Secondary)	
Method	E_t (cm)	E_R (°)	E_t (cm)	E_R (°)
LCCNet	1.01	0.12	52.5	1.47
CMRNet	1.57	0.10	52.9	1.49
CMRNext	1.89	0.08	7.07	0.23
Stage-1 (N=100)	2.63	0.187	6.23	0.137
Ours (N=100)	0.89	0.038	4.97	0.030

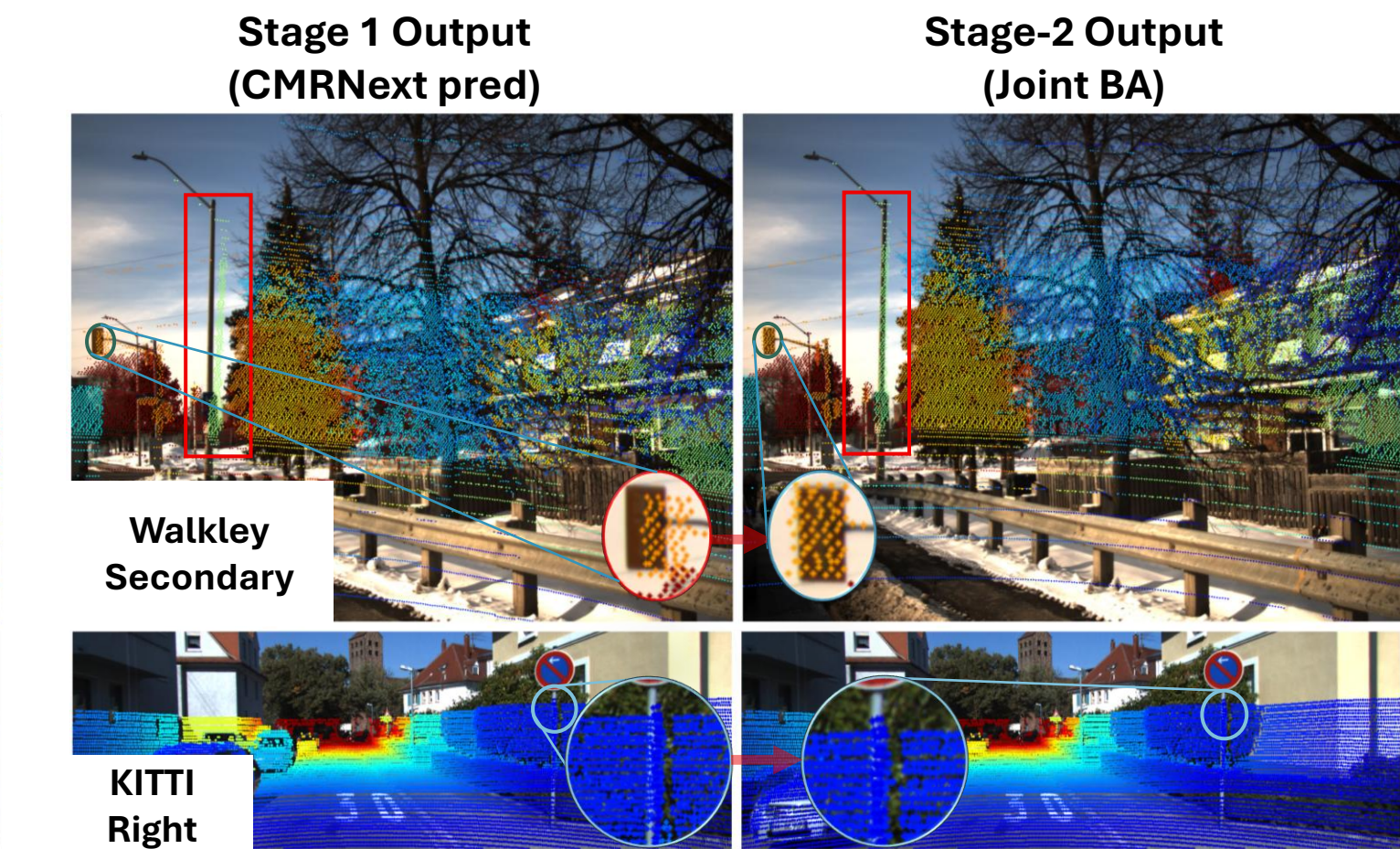


Conclusion

We introduced a two-stage framework that converts learned pairwise camera-LiDAR predictions into a globally consistent multi-camera calibration. The method improves both absolute calibration and inter-camera consistency, with especially strong gains when single-camera predictions are less reliable.

- Walkley Dataset:**
 - The Stage-2 joint bundle adjustment successfully recovers from poor single-camera initializations.
 - Reduced secondary-camera translation error from **108.6 cm/ 1.66° to 3.14 cm/ 0.20°**.
 - Improved inter-camera translation consistency from **23.9 cm to 21.0 cm** compared to reprojection-only BA.
 - Shows that the relative-pose prior helps stabilize calibration on out-of-domain data.

Walkley	Left Camera (Primary)		Left Camera (Secondary)		IC
Method	E_t (cm)	E_R (°)	E_t (cm)	E_R (°)	Δt (cm)
Stage-1	87.2	1.09	108.6	1.66	87.0
BA (reproj. only)	25.7	0.83	3.57	0.23	23.9
Ours	22.6	0.73	3.14	0.20	21.0



Future Work

- Extend to rigs with more than two cameras
- Explore online receding-window bundle adjustment.
- Investigate end-to-end training with the geometric refinement objective.

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Contact:

- Aziz Al-Najjar
- Data Science, Analytics and AI
- Carleton University, Ottawa, Canada
- Email: AzizAlNajjar@cmail.carleton.ca

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